PROJECT- BOARD GAME PLAYING AGENT

Team - 09

1. Krishna Vamsi - 2520030344
2. NV.Karthikeya - 2520030307 , 3.M.Nishant - 2520030323

INTRODUCTION

A Board Game Playing Agent is an AI that plays turn-based games by exploring possible future moves using the Minimax algorithm with Alpha-Beta Pruning. Its performance depends on search depth and heuristic evaluation functions that score board positions. This project helps analyze how these factors influence the agent's decision-making and overall gameplay effectiveness.

ABSTRACT

This project involves designing and implementing an AI agent capable of playing a turn-based board game (such as Tic-Tac-Toe or Connect Four) using classical game tree search algorithms. The agent uses Minimax search with Alpha-Beta Pruning to explore possible future game states and make optimal decisions. Students experiment with varying search depths to observe the trade-off between computation time and move quality, and design custom heuristic evaluation functions to estimate board positions when the full game tree cannot be explored.

Objective

Objective: To design and implement an AI-powered agent capable of playing a turn-based board game by applying the Minimax algorithm with Alpha-Beta Pruning for optimal decision-making. The agent will explore possible game states up to a defined search depth and use heuristic evaluation functions to score non-terminal board positions. The primary goal is to analyze how varying the search depth and heuristic strategies affect the agent's performance, accuracy, and computation time. This project aims to provide hands-on understanding of adversarial search techniques and their real-world application in game-playing AI systems.

THANK YOU